CAPSTONE PROJECT

SMART COMPRESSION SLEEVE

For Medical Recovery: Surgical & Lymphatic Support

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Group #8

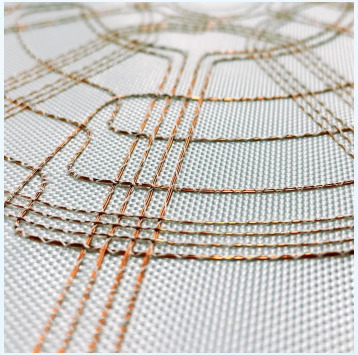
Introduction

Problem

Current compression sleeves provide no real-time feedback, and users often apply too little or too much compression.

Importance

Improper compression can slow healing, reduce comfort, and increase health risks. Users rely on guesswork rather than being provided with data. Current sleeves are typically bulky, expensive, and not practical for everyday use.



Significance

The Smart Compressions Sleeve provides real time feedback using textile sensors ensuring compression stays within a safe and effective range. With the use of data, recovery becomes quicker and more comfortable.

Scope of Solution

3 Main Systems: Textile Pressure Sensing Layer, Embedded Electronics for wireless communication, and a User Interface

-Sensing: The pressure sensors must be flexible and capable of operating in a low-pressure range that is appropriate for ideal compression.

-Feedback: The system needs to provide real-time feedback (via app or on-sleeve indicators) to prevent over-compression or under-compression

Background (Competitive Analysis)

-Traditional Sleeves (e.g., Nike, Copper Fit): Accessible and portable but rely solely on "fit and feel" with no data feedback.

-Pneumatic Systems (e.g., NormaTec): Programmable and controlled but are bulky, expensive, and stationary.

-Fitness Wearables (e.g., Fitbit): Monitor general recovery metrics but cannot measure or adjust specific compression levels.

Capstone Specific Objectives

-Project Scope: Focus is strictly on sensor selection/validation, hardware assembly, BLE communication, and a proof-of-concept mobile app.

-Differentiation: Combine the portability of a traditional sleeve with the intelligent safety feedback of a medical device.

Implementation Strategy

Sensing Layer (Textile Integration)

- Utilize pressure sensors mapped around the sleeve to create a pressure distribution map.
- Sensors will be integrated or tucked underneath the fabric and connected to the microcontroller via conductive thread to maintain flexibility.

Embedded Electronics (Hardware)

- Microcontroller: ESP32-C3 Dev Module selected for native Bluetooth (BLE) capabilities, low cost, and ease of programming.
- Power: 500mAh LiPo battery with a dedicated charging board for safe USB recharging
- Alerts: Integration of haptic feedback vibration or LED lights to physically alert the user of improper placement or unsafe pressure.

Software & Logic

- Firmware: Wire firmware to read sensor data and implement BLE (Bluetooth Low Energy) communication protocols.
- Interface: A simple smartphone app to display real-time pressure readings and log history.

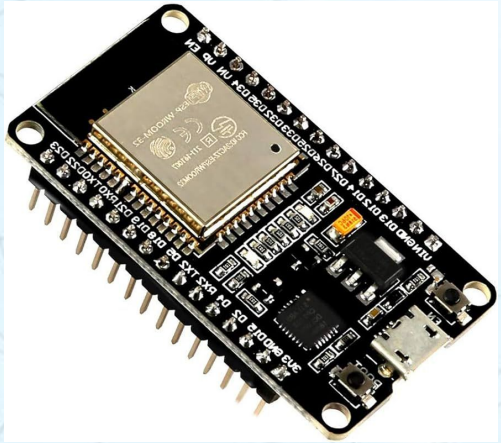
Materials & Budget

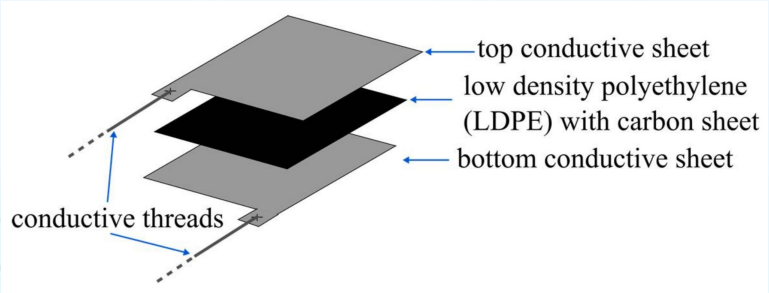
MCU: 2x ESP32-C3 Dev Modules ($16.00 total)

Sensors: 8x Flexible Textile Pressure Sensors ($80.00 total) — *Primary cost driver*

Power: 2x 500mAh LiPo Batteries + Charging Boards ($27.00 total)

Materials: Compression Sleeves, Conductive Thread, Haptic Motor ($53.00 total)

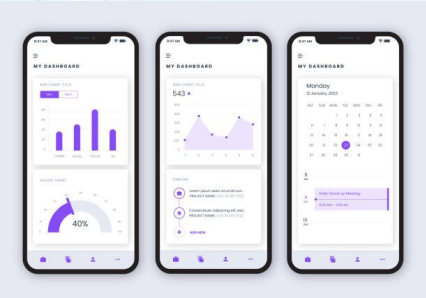




Prototype & Verification

Prototype Deliverable

- A "V1 Prototype" consisting of a functional compression sleeve with integrated sensors and a sleek electronics module.
- Wireless communication to a basic mobile app UI displaying real-time pressure status.



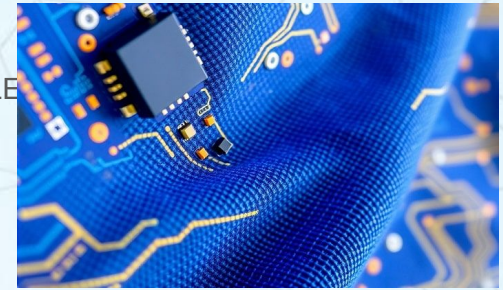
Verification & Testing (Test Matrix)

- **Sensor Calibration:** Develop a protocol to ensure pressure readings are accurate within therapeutic ranges
- **Connectivity:** Verify stable BLE data transmission from sleeve to app.
- **Durability:** Test battery life and textile durability under use conditions.

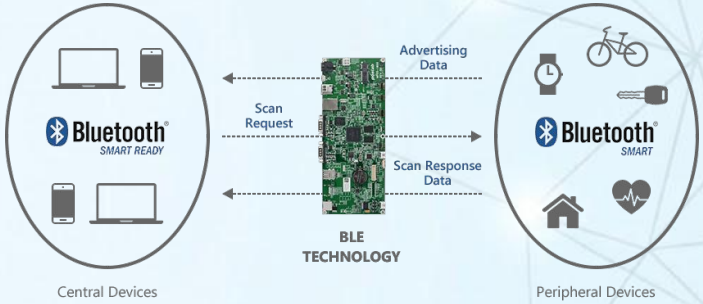
Success Criteria

- The device successfully alerts the user (physically or digitally) when compression levels are unsafe or ineffective.
- Demonstration the application of smart textiles to solve the "blind usage" problem of traditional compression

TASKS & TIMELINES



- Initial research and requirements, sensor selection and integration, electronics and firmware, BLE communication, mobile app and user interface creation, calibration and testing, and prototype assembly along with a working demonstration.
- **Critical Path:**
  Sensor Integration→Firmware Readout→BLE Link→App Feedback Loop→Calibration/Testing.
- **Milestones:**
  - 1 - Choose a sensor and integrate it seamlessly into sleeve.
  - 2 - Guarantee microcontroller reads sensors reliably, ensuring stable data.
  - 3 - BLE sends live data to mobile phone through application.
  - 4 - App displays the correct pressure state and alerts.
  - 5 - Complete calibration
  - 6 - Final prototype, report, and presentation
- **Build Components:**
  - Component procurement - All
  - Sensor/Textile Integration and protocol - Micah
  - Firmware/Electronics - Jack
  - BLE + App UI - Jose
  - Testing and documentation - All
- **Risks & Contingency:**
  - Sensor drift/inconsistent readings - simplify the zones (low/ok/high) and frequent re-zeroing.
  - Sensor placement and discomfort/failure - use a modular sensor patch or use protective encapsulation.
  - BLE instability - Send averaged updates at a lower rate
  - Battery life/high power draw - Reduce the sampling rate, utilize sleep mode when not needed.



MANAGEMENT & ASSESSMENTS

Tasks Organization:

- Research and design
- Hardware prototyping
- Software development
- Integration and testing
- Final Deliverables

Task Tracking:

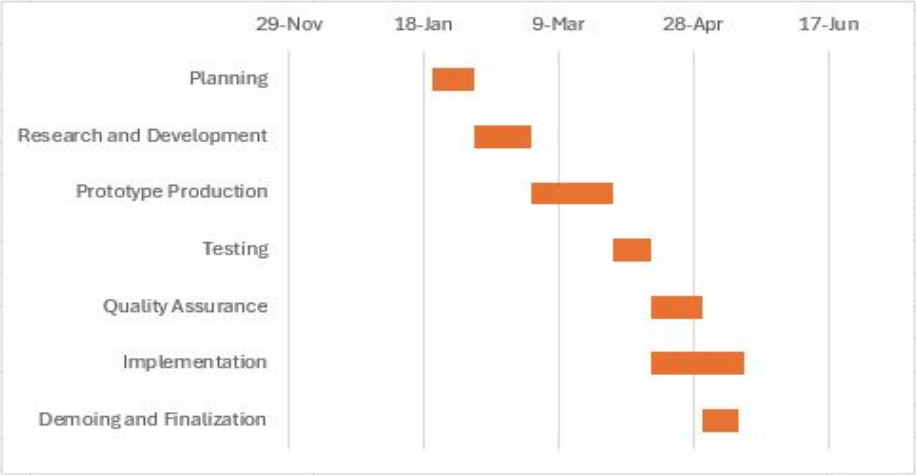
- Utilize a weekly tracking method where we mark a task as “Not started/In progress/Done”
- Milestone check-ins w/ team members on a consistent basis
- Gantt Chart

Test List:

- Sensor repeatability and drift check
- Calibration accuracy on pressure
- BLE latency & dropouts
- App display functionality
- Battery life and durability

Schedule - Gantt Chart

Task(s)			
Start Date	End Date	Description	Duration (Days)
21-Jan	6-Feb	Planning	16
6-Feb	27-Feb	Research and Development	21
27-Feb	29-Mar	Prototype Production	30
29-Mar	12-Apr	Testing	14
12-Apr	1-May	Quality Assurance	19
12-Apr	15-May	Implementation	35
1-May	15-May	Demoing and Finalization	14



Organization - Function Based

Hardware and Sensors

- Sensor selection and validation
- Sensor Integration to sleeve
- Power System and Battery Management
- Calibration and durability testing

Firmware and Wireless Connection

- Microcontroller setup and data obtaining
- Bluetooth Communication
- Power optimization and data formatting
- Hardware-Software integration support

User Interface and Application

- Mobile App / User Interface Design
- Real-time pressure display and alerts
- Data logging and basic visualization
- User experience and system usability

Shared Responsibilities

- System integration
- Documentation and reporting
- Presentation preparation and final demonstration

REFERENCES

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THANK YOU!